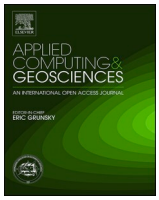
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Advancing geological image segmentation: Deep learning approaches for rock type identification and classification

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ABSTRACT

This study aims to tackle the obstacles linked with geological image segmentation by employing sophisticated deep learning techniques. Geological formations, characterized by diverse forms, sizes, textures, and colors, present a complex landscape for traditional image processing techniques. Drawing inspiration from recent advancements in image segmentation, particularly in medical imaging and object recognition, this research proposed a comprehensive methodology tailored to the specific requirements of geological image datasets. To establish the dataset, a minimum of 50 images per rock type was deemed essential, with the majority captured at the University of Las Palmas de Gran Canaria and during a field expedition to La Isla de La Palma, Spain. This dual-source approach ensures diversity in geological formations, enriching the dataset with a comprehensive range of visual characteristics. The study involves the identification of 19 distinct rock types, each documented with 50 samples, resulting in a comprehensive database containing 950 images. The methodology involves two crucial phases: initial preprocessing of the dataset, focusing on formatting and optimization, and subsequent application of deep learning models—ResNets, Inception V3, DenseNets, MobileNets V3, and EfficientNet V2 large. Preparing the dataset is crucial for improving both the quality and relevance, thereby to ensure the optimal performance of deep learning models, the dataset was preprocessed. Following this, transfer learning or more specifically fine-tuning is applied in the subsequent phase with ResNets, Inception V3, DenseNets, MobileNets V3, and EfficientNet V2 large, leveraging pre-trained models to enhance classification task performance. After fine-tuning eight deep learning models with optimal hyperparameters, including ResNet101, ResNet152, Inception-v3, DenseNet169, DenseNet201, MobileNet-v3-small, MobileNet-v3-large, and EfficientNet-v2-large, comprehensive evaluation revealed exceptional performance metrics. DenseNet201 and InceptionV3 attained the highest accuracy of 98.49% when tested on the original dataset, leading in precision, sensitivity, specificity, and F-score. Incorporating preprocessing steps further improved results, with all models exceeding 97.5% accuracy on the preprocessed dataset. In K-Fold cross-validation (k = 5), MobileNet V3 large excelled with the highest accuracy of 99.15%, followed by ResNet101 at 99.08%. Despite varying training times, models on the preprocessed dataset showed faster convergence without overfitting. Minimal misclassifications were observed, mainly among specific classes. Overall, the study's methodologies yielded remarkable results, surpassing 99% accuracy on the preprocessed dataset and in K-Fold cross-validation, affirming the efficacy in advancing rock type understanding.

1. Introduction

In geological studies, the accurate segmentation and categorization of rock types from complex image datasets are paramount for understanding geological formations. The intricate nature of geological formations, characterized by diverse forms, sizes, textures, and colors, poses a formidable challenge for traditional image processing techniques. To address these obstacles, this study draws inspiration from

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recent progress in image segmentation, particularly in domains such as medical imaging and object recognition. Geological image segmentation holds a crucial role in comprehending and categorizing various rock types, posing challenges owing to the complex nature of geological formations. This research line addresses the challenges associated with geological image segmentation by employing advanced deep-learning methodologies.

It is possible to find references, which show the importance of this topic and justify the current proposal. The related works encompasses a spectrum of studies addressing image segmentation, deep learning frameworks, and the application of specific models across various domains. Faska et al. (2023) propose a robust ensemble-learning framework for image segmentation, providing potential insights into improving segmentation performance in geological datasets. Khened et al. (2021) contribute a generalized deep learning framework for whole-slide image segmentation, aligning with the broader theme of segmentation tasks and offering optimization strategies. Kromp et al.'s (2019) deep learning architectures for immunofluorescence-based nuclear image segmentation may inspire innovative approaches for rock type segmentation challenges. Cui et al.'s (2020) unified framework for low-shot medical image segmentation with scarce data provides techniques for effective segmentation with limited data, applicable to the constraints of geological datasets. Additionally, studies by Salim et al. (2023); Ogundokun et al. (2023); Winoto et al. (2020); Yuan et al. (2022) focus on model-specific considerations, lightweight models, and efficiency, providing valuable insights applicable to geological image segmentation. Wang et al. (2018) highlight interactive medical image segmentation using deep learning with image-specific fine-tuning, providing insights into the fine-tuning approach discussed in geological image segmentation. Drawing insights from medical imaging, Kamraoui et al. (2022) showcase DeepLesionBrain's broader generalization, which may inspire domain-specific improvements in geological image segmentation. Tahir et al. (2023) present DSCC_Net's multi-classification deep learning models for diagnosing skin cancer, offering considerations for specific medical applications. Jha and Pant (2021) offer perspectives on stochastic gradient descent and discriminative fine-tuning on various neural networks, including ResNet, DenseNet, Inception-ResNet, and MobileNet. Wang et al. (2020) present a novel image classification approach via Dense-MobileNet models, contributing comparisons and adaptations relevant to geological image classification. Rahman et al.'s (2023) deep CNN approach for detecting and classifying local fruits aligns with the image classification aspect of the current research. Minaee et al., 2021 survey on image segmentation using deep learning offers a broad understanding of methodologies, guiding the selection and optimization of segmentation techniques in the geological context. Xu et al. (2022) explore closing the generalization gap in federated medical image segmentation, aligning with challenges faced in geological datasets. Jain et al. (2021) discusses the lightweight multi-label image classification model based on Inception modules provides insights into creating efficient models. Adaptations from this work may be explored for optimizing the geological image classification process. Rabbi et al. (2022) discusses transfer learning for autism spectrum disorder detection, suggesting versatile models for geological image segmentation. Moreover, the studies of Sadati and Rezaie (2021); Zailan et al. (2022); Xavier (2024); Sarker et al. (2020); Makris et al. (2020); Tomar (2020) contribute insights into feature extraction, transfer learning, and deep learning applications in diverse domains, providing valuable perspectives for geological image segmentation challenges. Saleem et al. (2020) explore deep learning for plant disease classification. Their work sheds light on using convolutional neural networks (CNNs) for image recognition tasks, potentially applicable to geological formations analysis.

According to this state-of-the-art, this proposal tackles these challenges through a comprehensive methodology that utilizes cutting-edge deep learning models. The conceptual framework draws inspiration from studies such as Faska et al. (2023), which introduced a robust ensemble-learning framework designed for image segmentation. Khened et al. (2021) contributed a comprehensive deep-learning framework for whole-slide image segmentation. Leveraging insights from Kromp et al. (2019); Cui et al. (2020) delve into deep learning architectures for nuclear image segmentation and unified frameworks for low-shot medical image segmentation, respectively. The methodology applied aims to adapt these techniques to the geological context. In response to the limited dataset, a crucial aspect of our methodology involves data augmentation, where a novel approach of image tiling or segmentation is employed to address the scarcity of images in individual classes. This technique, inspired by studies on efficient lightweight models (Ogundokun et al., 2023; Winoto et al., 2020; Yuan et al., 2022) expanded the dataset size by quadrupling it, achieved through the segmentation of large images into smaller, more manageable tiles. Normalization and resizing, integral steps in data preprocessing, are influenced by the works of Wang et al. (2018) and Kamraoui et al. (2022), emphasizing the importance of these techniques in enhancing model generalization and performance.

This study strategically applies normalization based on model architecture and selectively applies resizing to meet Convolutional Neural Network (CNN) input size criteria, as detailed in the methodology. The dataset, carefully curated to represent a comprehensive range of geological formations, underwent a meticulous partitioning strategy following insights from Tahir et al. (2023), who emphasize the importance of K-fold cross-validation for robust model evaluation. This approach safeguards against overfitting and ensures a reliable assessment of model performance on entirely novel and unseen data. The implementation of transfer learning, influenced by the work of Jha and Pant (2021), allows us to leverage pretrained models such as ResNets, Inception-v3, DenseNets, MobileNets-v3, and EfficientNet-v2-large. The fine-tuning process, guided by optimal hyperparameters determined through a grid search approach, aligns with the methodology proposed by Wang et al. (2020); Rahman et al. (2023). The comprehensive results highlight the success of our methodologies, validated through K-Fold cross-validation and performance metrics such as accuracy, F-score, precision, sensitivity, and specificity. The robustness of the models, as indicated by the training accuracy and loss graphs and the confusion matrices, underscores the efficacy of the proposed approach in advancing the understanding and categorization of rock types through geological image segmentation.

Geological image segmentation plays a crucial role in various fields, including geological mapping, mineral exploration, and petrology. The proposal highlights the significance of deep learning techniques in enhancing the accuracy and efficiency of geological image segmentation. By employing deep learning models, geologists can effectively identify and classify rock types from images, leading to improved decision-making and resource management. This paper's contributions are essential for advancing geological research and addressing societal challenges related to resource exploration and sustainable development.

The paper will be structured as follows: Section 2 will talk about the materials and methods; Section 3 will showcase the results of the experiment and spark the discussion that will eventually lead to the conclusions that will be showcased in Section 4.

2. Methodology

The research methodology employed in this study revolves around a clear conceptual framework, primarily aimed at tackling a classification objective. The methodology comprises two essential phases: preprocessing of the dataset and the application of deep learning models. The initial phase of the methodology concentrates on dataset preprocessing, a critical step to ensure the data is appropriately formatted and optimized for subsequent analysis. The preprocessing stage plays a vital role in improving the quality and pertinence of the dataset, thereby enhancing the overall effectiveness of the deep learning models. After completing the dataset preprocessing, the next stage involves

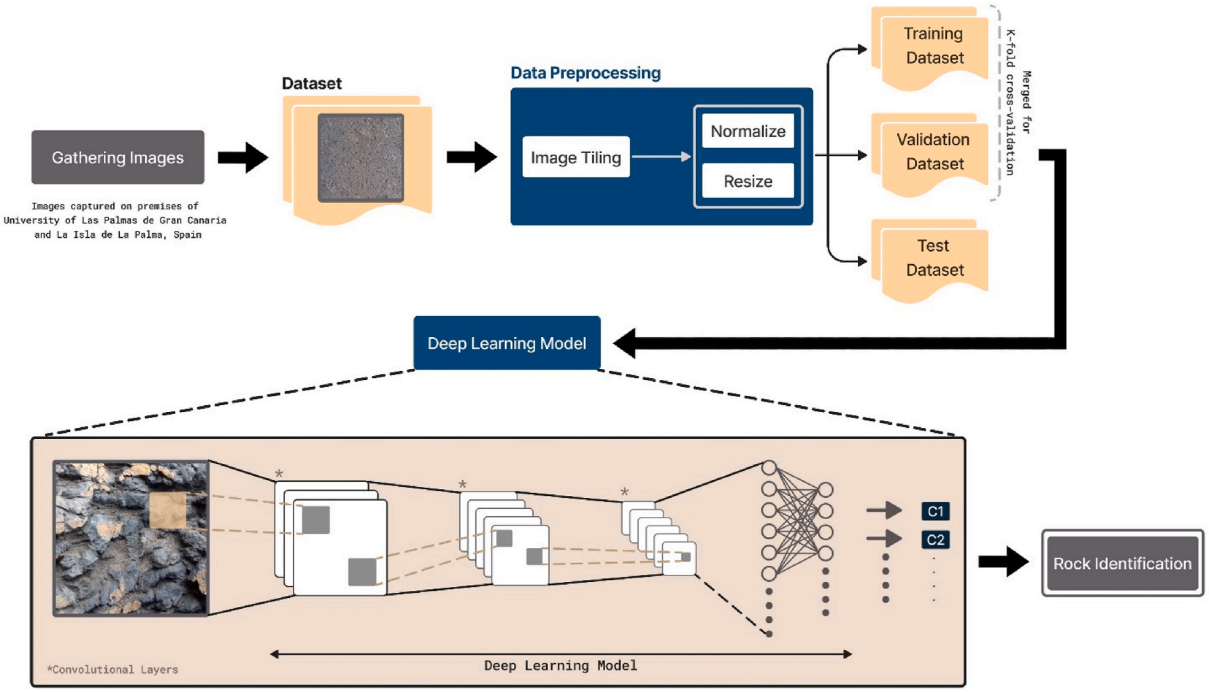


Fig. 1. Overview of complete Methodology workflow.

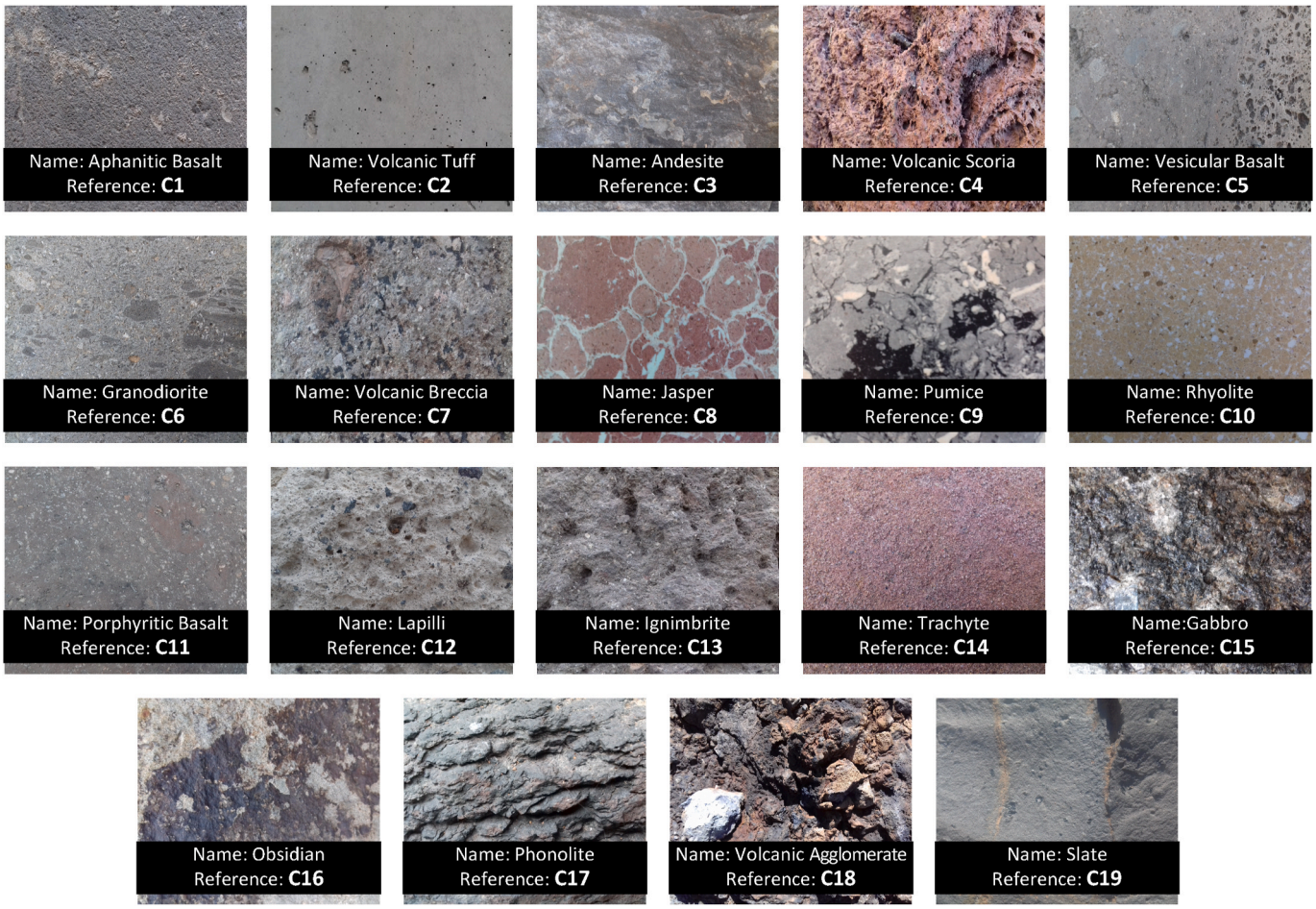


Fig. 2. Reference image from each class within the dataset.

Table 1
Class names and referenced name for dataset labeling.

Class Name	Referenced Name	Class Name	Referenced Name
Aphanitic Basalt	C1	Porphyritic Basalt	C11
Volcanic Tuff	C2	Lapilli	C12
Andesite	C3	Ignimbrite	C13
Volcanic Scoria	C4	Trachyte	C14
Vesicular Basalt	C5	Gabbro	C15
Granodiorite	C6	Obsidian	C16
Volcanic Breccia	C7	Phonolite	C17
Jasper	C8	Volcanic Agglomerate	C18
Pumice	C9	Slate	C19
Rhyolite	C10		

implementation of transfer learning or more specifically fine-tuning. Specifically, ResNets, Inception V3, DenseNets, MobileNets V3, and EfficientNet V2 large are employed and fine-tuned to the dataset. Transfer learning utilizes pre-trained models on extensive datasets known as IMAGENET dataset and enables the transfer of knowledge acquired from one task to another, thereby improving the performance of the models in the current classification task. The complete methodology workflow as well as the fundamental architecture of deep learning models utilized in the study is depicted in Fig. 1, providing a clear and concise overview of the two-phase process.

2.1. Dataset characterization

The complexity associated with identifying rocks stems from the diverse nature of geological formations, encompassing a range of forms, sizes, textures, and colors. Differentiating among these varied rock types poses a significant challenge for those without specialized training. A thorough review of existing literature on rock identification emphasizes the crucial need for a comprehensive dataset. Typically, such datasets consist of images, occasionally supplemented by additional features like chemical composition. This combination of visual and supplementary information is vital for fostering a more accurate and nuanced understanding of the intricate nuances involved in categorizing rocks. In establishing the dataset, it was determined that a minimum of 50 images per rock type is essential for effective identification. The majority of these images were captured on the premises of the University of Las Palmas de Gran Canaria and during a field expedition to La Isla de La Palma, both situated in Spain. In particular, the Canary Islands are a Spanish archipelago located in the Atlantic Ocean, off the northwestern coast of Africa. They are situated approximately 100 km (62 miles) west of Morocco and the Western Sahara. The islands comprise seven main islands and several smaller islets, all of volcanic origin. The Canary Islands lie between 27° and 29° north latitude and span from 13° to 18° west longitude.

This dual-source approach ensures a diverse representation of geological formations, enriching the dataset with a comprehensive array of visual characteristics. The process of rock identification in this study involves a broad and varied dataset, crucial for ensuring the precision of subsequent analysis and classification. A total of 19 distinct rock types were identified, each documented with 50 samples, resulting in a comprehensive database containing 950 images. These images are uniformly formatted as JPEG (Joint Photographic Experts Group), ensuring compatibility and ease of use. The acquisition of the rock images has been done by Geologists, under their expert consideration in the process of the image acquisition. All images were acquired at an approximate distance of 10–15 cm using a 12-megapixel camera. The subjects were large stones, allowing for a focused analysis of texture without background interference. To maintain the original scale and resolution, no post-processing modifications were applied to the images. The captures were conducted outdoors under natural daylight conditions. For illustrative purposes, Fig. 2 presents the reference image from each class within the dataset, providing a visual anchor.

The Canary Islands are a volcanic archipelago comprising diverse rock types, including basalts (dark-colored, fine-grained from fluid lava flows), trachytes (lighter-colored, glassy from less fluid lava), and phonolites (lightest, very fine-grained from viscous, silica-rich lava). These rocks showcase the islands' volcanic history and influence their landscapes. Besides, granite rock types have been included because they are the most used in the world, so the dataset has more variability due to the effect of globalization.

2.2. Data labelling

For improved clarity in understanding and effortless cross-referencing, each class in the methodology has been assigned a unique nomenclature. The relationship between the dataset class names and the

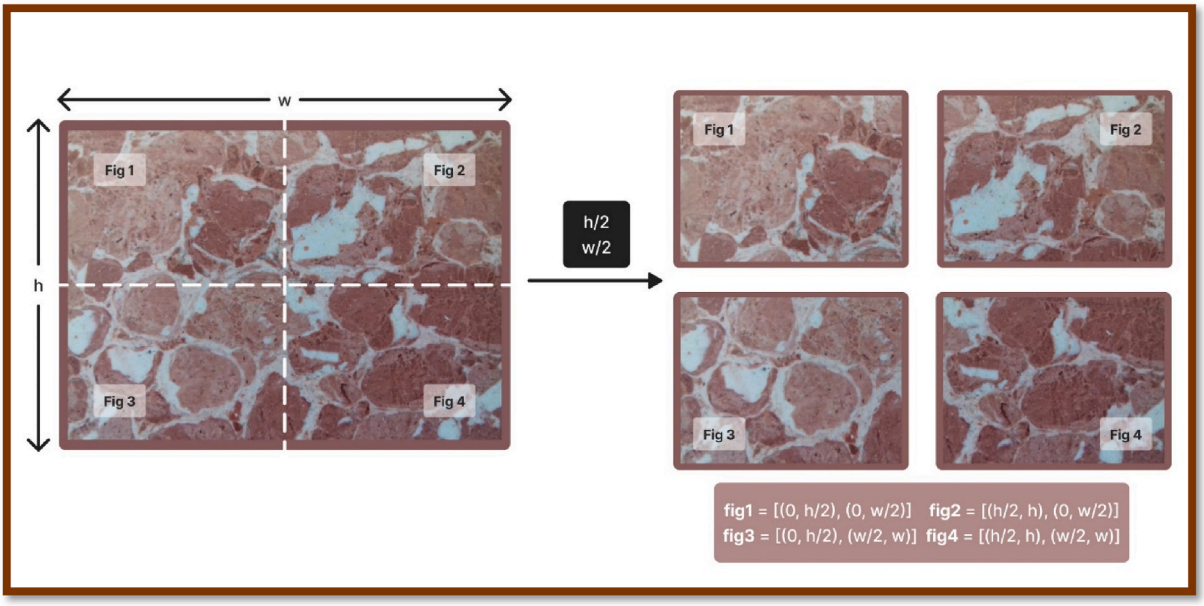


Fig. 3. Showing the Process of Image Tiling adopted in this Research.

names referenced in the study is detailed in Table 1.

2.3. Data preprocessing

Preprocessing plays a central and widely adopted role in preparing datasets prior to training deep learning models. It entails a set of methodologies crafted to refine datasets, ensuring their quality, and facilitating the extraction of accurate and meaningful insights. The efficacy of data preprocessing is crucial in readying datasets for model training, contributing significantly to improved model performance. One indispensable technique in the realm of data preprocessing is data augmentation. This pivotal approach involves applying various transformations to the existing training dataset, introducing artificial diversity to the data. Beyond merely expanding the dataset, data augmentation serves a dual purpose. Firstly, it improves the model's capacity to generalize adeptly to instances it hasn't encountered before, useful for real-world application. Secondly, it fosters robustness by exposing the model to a broader spectrum of scenarios. By strategically expanding the dataset through data augmentation, deep learning models are not only better equipped to handle the intricacies and variations inherent in real-world data but also demonstrate an enhanced capability to address diverse challenges. These preprocessing and augmentation processes plays a significant role in elevating overall model performance and reliability.

2.3.1. Image tiling

The dataset comprises a diverse set of 19 classes, each characterized by a limited number of images, posing a challenge in developing high-performing models. To overcome this limitation, data augmentation becomes imperative. This involves applying various transformations to the existing training dataset, introducing artificial diversity. While foundational techniques like rotation, flip, and brightness adjustments are commonly used, the unique characteristics of the dataset in this study, where multiple classes share similar features, demand a more nuanced approach. Conventional techniques may not yield significant improvements. Given the dataset's large image resolutions or dimensions, a novel strategy was employed: image tiling or segmentation. This technique divides large images into smaller, more manageable tiles or patches. Notably, the average height and width of images in the dataset were approximately 3000 $\times$ 2000 pixels, which is notably large when compared with the input size ranging from 224 to 299 pixels in the CNNs employed in the study. To augment the dataset, images were evenly split both horizontally and vertically, resulting in the creation of four distinct images from each original one, with each quadrant now a new image. This innovative method substantially increased the dataset's image count fourfold. Fig. 3 offers an intricate illustration of this segmentation strategy. This method not only addresses the scarcity of images in individual classes but also contributes to a more extensive and diverse dataset as no two quadrants images are same, therefore essentially creating a new dataset from the original images.

2.3.2. Image normalization

Normalization emerges as another vital data preprocessing technique utilized in this investigation. In this scenario, we standardize a tensor image by employing its mean and standard deviation (std). This procedure guarantees that the input features maintain consistent scaling and centralization, thereby fostering enhanced convergence during the training process. The selection of appropriate mean and standard deviation values, representing the 1st and 2nd order statistics per channel, allows for the calculation of the z-score of our data channel by channel. This normalization technique, often referred to as 'standardization' (Saber, A et al., 2021), involves applying the following equation for calculation:

$$output_{channel} = \frac{input_{channel} - mean_{channel}}{std_{channel}} \quad (1)$$

where $input_{channel}$ represents the input data, $mean_{channel}$ signifies the mean of the data, and $std_{channel}$ denotes the standard deviation of the data. Incorporating this method ensures the standardization of the data, thereby contributing to the overall efficacy of the data preprocessing process in bolstering model training. The normalization process plays a pivotal role in improving the model's generalization across diverse types of images. By ensuring that each channel in the image data adheres to a standard normal distribution, this process significantly aids in improving the model's capacity to generalize effectively across a wide range of images. In this study, the application of normalization was tailored based on model architecture and other pertinent factors, with the decision not to uniformly apply it to all models. This nuanced approach allows for model-specific considerations, acknowledging the potential variations in the impact of normalization across different architectures and scenarios.

2.3.3. Image resizing

To meet the specified input size criteria of Convolutional Neural Network (CNN) models, a crucial preprocessing step involves resizing the acquired images. The difficulty of dealing with images of different dimensions, a common occurrence in real-world situations, prompted the adoption of a data augmentation technique for the training dataset. Specifically, the Random Resized Crop technique was employed, randomly selecting a portion of the image and applying cropping and resizing operations to meet the specified input size requirements. This dynamic method played a significant role in enriching the diversity of the training data, consequently enhancing the robustness of the deep learning models. In contrast, the validation and testing datasets underwent a more straightforward resizing process. Images within these subsets were resized without additional augmentation, preserving the original content while conforming to the stipulated input dimensions of the CNN models. The resizing operation targeted three distinct sets of dimensions based on model depth, size, and parameters: 224 $\times$ 224, 256 $\times$ 256, and 299 $\times$ 299. These standardized sizes were carefully selected to align with the specific requirements of the CNN models utilized in this study, ensuring uniformity in input sizes across all models.

2.3.4. Data splitting

The dataset was meticulously partitioned into three distinct subsets with precision: 70% of the data is allocated for training, while 15% each is reserved for validation and testing purposes. This deliberate partitioning strategy was meticulously crafted to facilitate a thorough assessment of the model's performance at various stages of development. The training subset, comprising 70% of the dataset, serves as the foundational bedrock for the model, allowing it to assimilate and adapt to inherent patterns within the data. Simultaneously, the validation subset, constituting 15% of the dataset, plays a crucial role in providing an interim assessment during training, evaluating the model's generalization capabilities. The remaining 15% of the dataset is exclusively designated for the testing subset, which is a crucial aspect in evaluating the model's performance on unseen data, thus serving as a dependable indicator of the model's real-world applicability.

To alleviate the risk of overfitting, K-fold cross-validation was employed in the study. This technique involves dividing the amalgamated train and validation dataset, now comprising 85% of the dataset was divided into K folds or subsets and K times iteratively utilizing K-1 folds for training, while reserving the remaining fold for validation. Through systematic rotation of the folds, each data point is employed for validation exactly once. This approach not only tackles overfitting concerns, where the model might perform well on the training data but poorly on new, unseen data, but also improves the robustness of the model's performance estimation across various subsets of the dataset. Though the training and validation datasets were amalgamated for K-fold cross-validation, the originally created testing dataset, comprising 15% of the dataset, remained untouched. This method offers a succinct and accurate evaluation of each fold, with the separation ensuring a

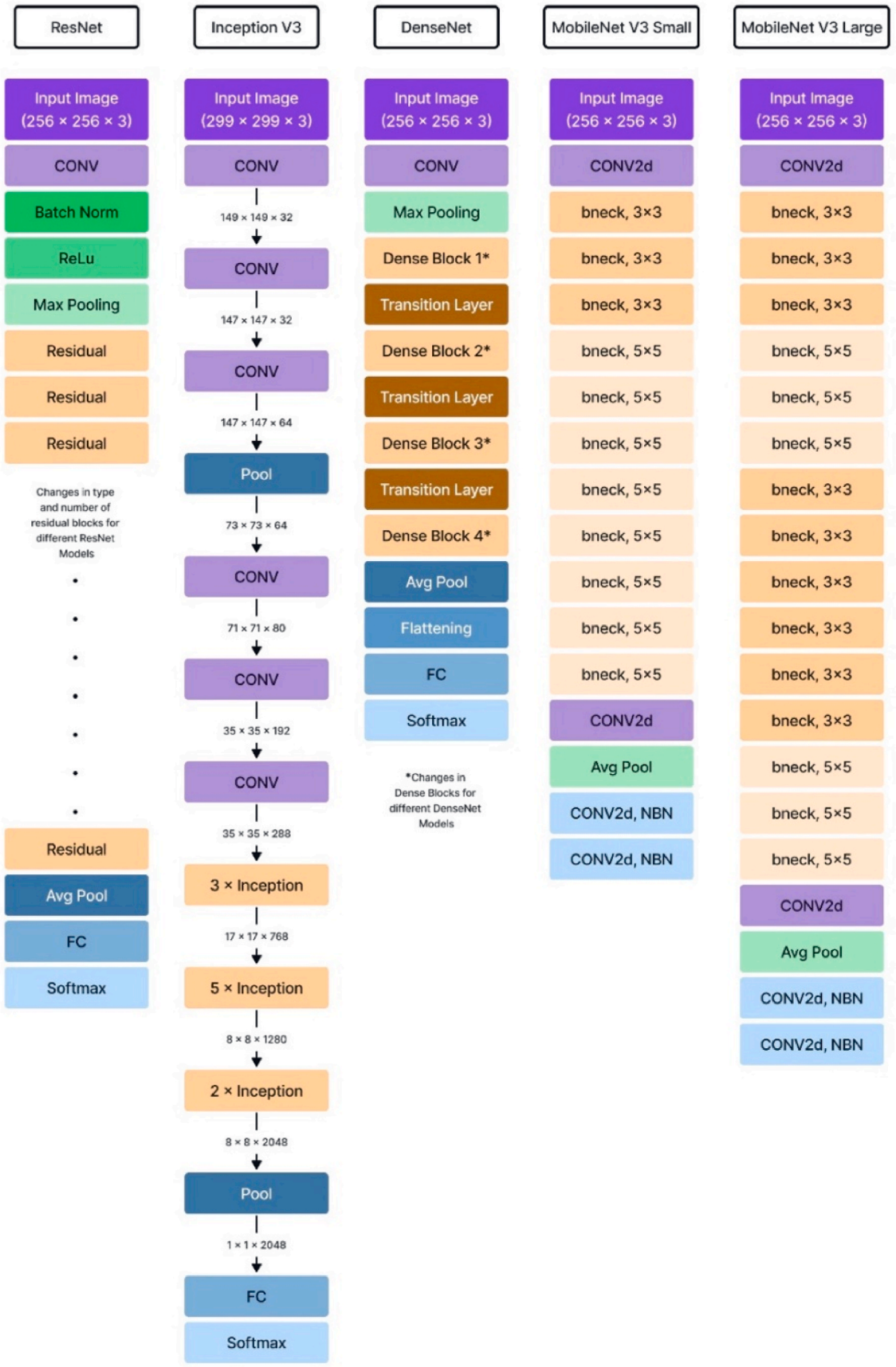


Fig. 4. ResNet, Inception V3, DenseNet, MobileNet V3 small and MobileNet V3 large architectures.

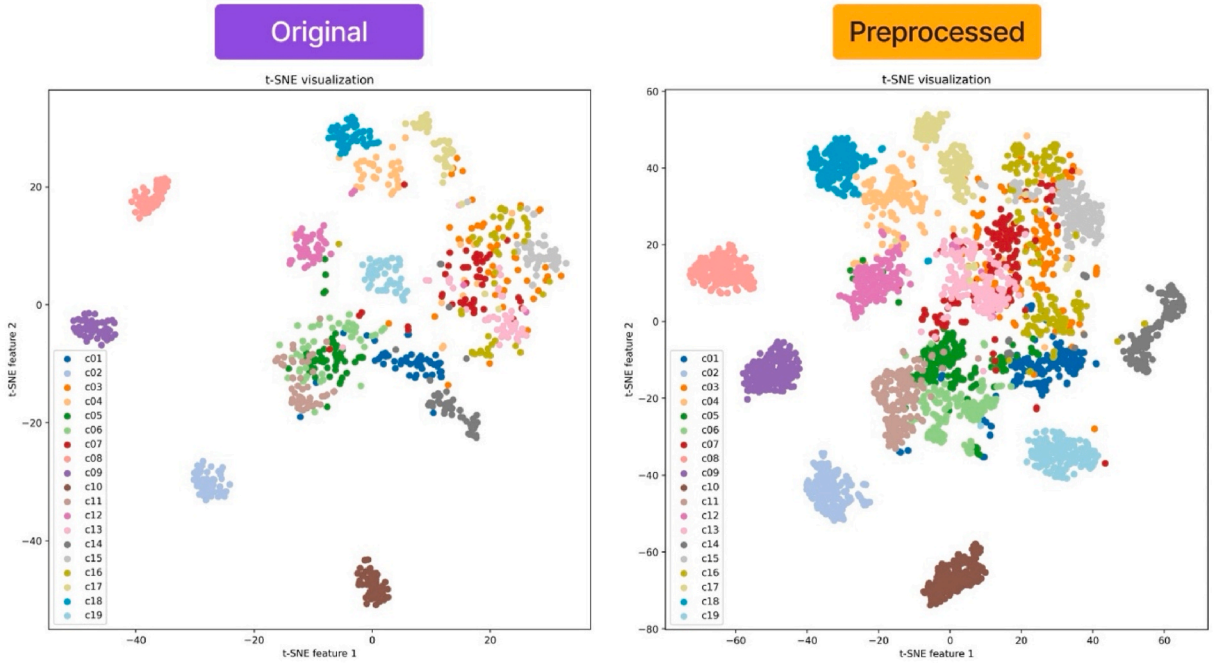


Fig. 5. t-SNE plot of the original and the preprocessed dataset.

Table 2
Basic Configurations and parameters for model for grid search.

Configuration/Parameters	Value
Optimizer	ADAM
Batch Size	16
Max Epoch	50
Learning Rate	0.001, 0.0001, 1e-05
Weight Decay	1e-07, 1e-08, 1e-09

Table 3
Accuracy received on all sets of parameters in grid search.

Learning Rate	Weight Decay	Accuracy (%)
0.001	1e-07	95.08
0.001	1e-08	95.96
0.001	1e-09	96.14
0.0001	1e-07	97.01
0.0001	1e-08	98.94
0.0001	1e-09	98.59
1e-05	1e-07	98.24
1e-05	1e-08	98.77
1e-05	1e-09	98.77

Table 4
Optimizer and parameter settings.

Sr. no	Parameter	Value
1	Optimizer	ADAM
2	Batch Size	16
3	Maximum Epoch	50/100
4	Learning Rate	0.0001
5	Weight Decay	1e-08

dependable and impartial assessment of the model's performance on entirely new and unseen data, thereby yielding a credible measure of model performance (Fushiki, T. , 2011).

Table 5
Performance comparison of various deep learning models on the original dataset.

Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-score (%)
ResNet101	97.74	98.02	97.74	99.87	97.73
ResNet152	96.99	97.51	96.99	99.83	96.95
Inception V3	98.49	98.68	98.49	99.91	98.48
DenseNet169	97.74	98.02	97.74	99.87	97.73
DenseNet201	98.49	98.68	98.49	99.91	98.48
MobileNet V3 small	94.73	95.39	94.73	99.70	94.79
MobileNet V3 large	96.99	97.42	96.99	99.83	97.02
EfficientNet V2 large	97.74	98.02	97.74	99.87	97.66

Table 6
Performance comparison of various deep learning models on the preprocessed dataset.

Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-score (%)
ResNet101	98.94	98.98	98.94	99.94	98.94
ResNet152	98.94	99.00	98.94	99.94	98.94
Inception V3	97.89	98.06	97.89	99.88	97.91
DenseNet169	99.12	99.17	99.12	99.95	99.12
DenseNet201	98.59	98.67	98.59	99.92	98.60
MobileNet V3 small	98.59	98.70	98.59	99.92	98.59
MobileNet V3 large	98.77	98.82	98.77	99.93	98.76
EfficientNet V2 large	99.12	99.17	99.12	99.95	99.12

2.4. Convolutional neural networks (CNN)

The rise of Convolutional Neural Networks (CNNs) in image processing has been extraordinary, largely due to their exceptional

Table 7
5-Fold cross validation performance of all deep learning models.

Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-score (%)
ResNet101	99.08	99.12	99.08	99.94	99.08
ResNet152	98.70	98.73	98.70	99.92	98.69
Inception V3	98.52	98.57	98.52	99.91	98.52
DenseNet169	98.73	98.79	98.73	99.92	98.73
DenseNet201	98.31	98.40	98.31	99.90	98.31
MobileNet V3 small	98.66	98.73	98.66	99.92	98.66
MobileNet V3 large	99.15	99.19	99.15	99.95	99.15
EfficientNet V2 large	98.91	98.96	98.91	99.93	98.90

Table 8
10-Fold cross validation performance of all deep learning models.

Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-score (%)
ResNet101	98.71	98.75	98.71	99.12	98.71
ResNet152	98.71	98.78	98.71	99.12	98.71
Inception V3	98.24	98.30	98.24	99.91	98.24
DenseNet169	98.77	98.81	98.77	99.95	98.77
DenseNet201	98.63	98.69	98.63	99.93	98.62
MobileNet V3 small	98.94	99.00	98.94	99.96	98.94
MobileNet V3 large	99.14	99.17	99.14	99.96	99.13
EfficientNet V2 large	98.70	98.76	98.70	99.93	98.69

performance in a variety of image-related tasks. CNNs, a subset of neural networks, are named for their use of convolutional layers, which are particularly adept at detecting local patterns in images. These networks typically include four principal layers: convolutional, max pooling, fully connected, and output layers, arranged sequentially. One of the most significant advantages of CNN architecture is its flexibility, allowing for customization to meet the needs of specific tasks. This adaptability is highlighted by CNN's dual functionality as both a feature extractor and a classifier. The fully connected layer classifies data based on high-level features extracted through the preceding convolutional and pooling layers. The CNNs utilized in this study are from ResNet family, DenseNet family, Inception V3, MobileNet family and EfficientNet V2 large (He, K., Zhang, X., Ren, S., & Sun, J., 2016).

ResNet or Residual Architecture by the Microsoft Research team is one of the pioneering advancements in deep neural networks. Developed to tackle the complexity encountered during the training of exceptionally deep neural networks, ResNet employs residual learning framework that subdues the issue of vanishing gradients, and all of them can also operate on a 256 × 256-input size (He, K., Zhang, X., Ren, S., & Sun, J., 2016). The core of the ResNet architecture is its residual blocks, which deviate from traditional architectures by using shortcut connections or skip connections, and with each block consisting of three convolution layers. These connections facilitate a more direct flow of gradients throughout the network, thereby alleviating the issue of vanishing gradients and making the training of extremely deep networks easier (He, K., Zhang, X., Ren, S., & Sun, J., 2016). The ResNet-101 architecture comprises of 101 layers, including 33 blocks or 99 convolution layers, 1 average-pooling layer, and 1 max-pooling layer and ResNet-152 architecture, with 152 layers, consisting of 50 block or 150 convolution layers, 1 average-pooling layer, and 1 max-pooling layer.

The **Inception V3** architecture, crafted by the Google research team, stands as another high-performing CNN architecture known for its exceptional accuracy. Its widespread adoption in transfer learning tasks stems from its capacity to capture a wide array of features and seamlessly adapt to various tasks. Inception V3 comprises 48 layers and operates on a 299 × 299-input size. It employs inception modules, which consist of multiple convolutional filters of varying sizes. This design choice enables the network to capture features across different scales, empowering it to effectively discern intricate patterns within images (Szegedy, C et al., 2016). The Inception V3 network has incorporated the RMSProp Optimizer. Additionally, it has reduced the number of parameters by restructuring large convolution layers, such as breaking down the traditional 7 × 7 convolution into three 3 × 3 convolutions. For instance, in the Inception part of the network, three traditional inception modules, each with 288 filters at a 35 × 35 grid, have been

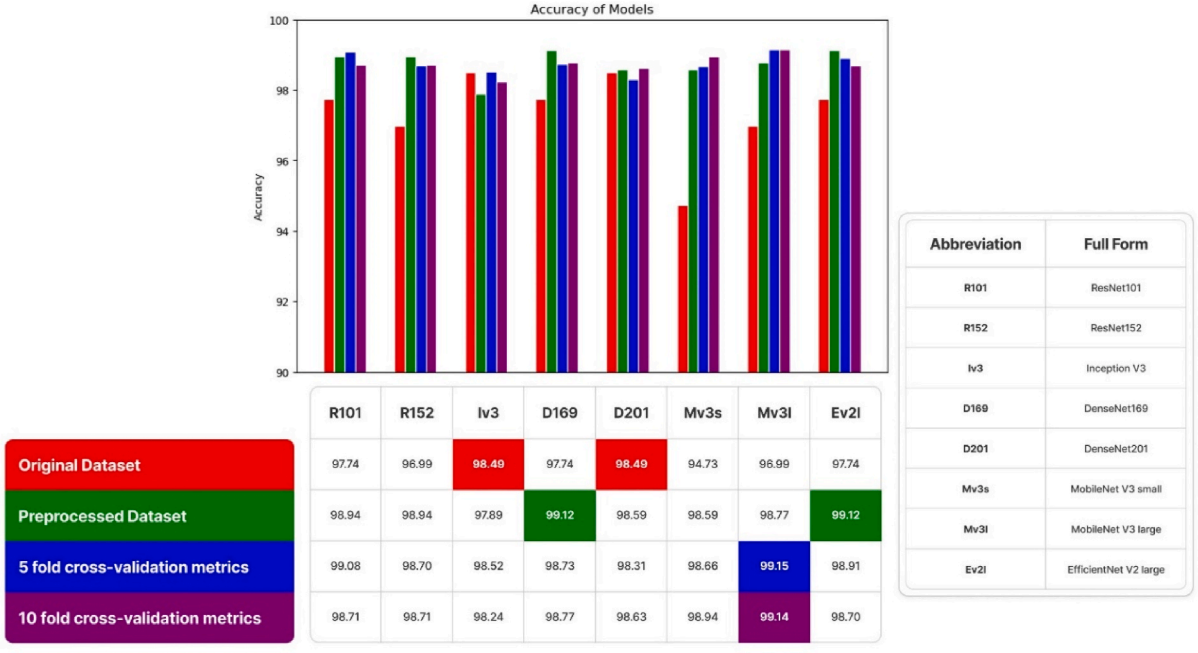


Fig. 6. Comparison of models performance across variations.

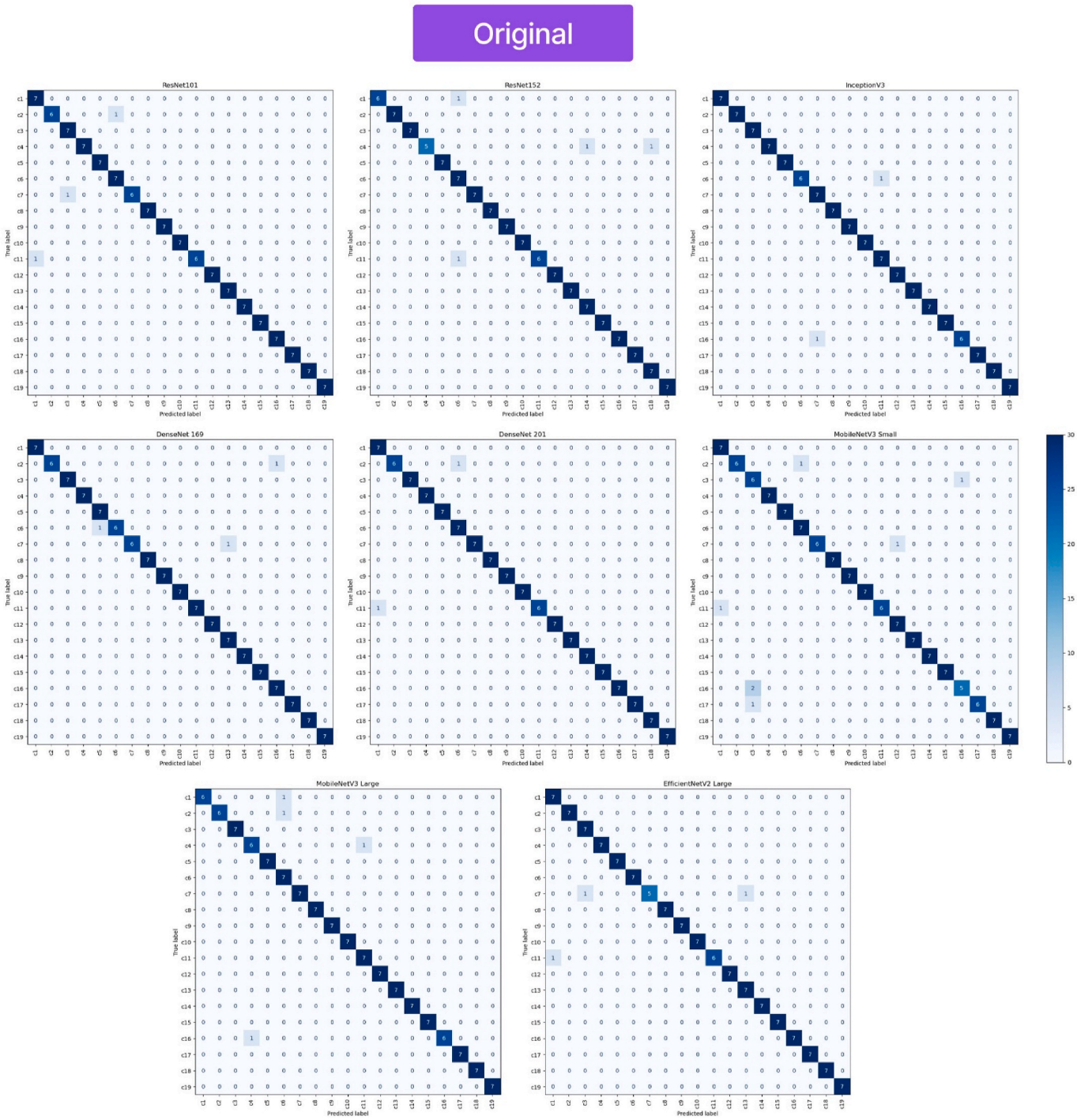


Fig. 7. Confusion matrix for all models on original dataset.

condensed to a 17 × 17 grid with 768 filters utilizing grid reduction techniques and other optimization methods. These modifications to the fundamental structure of the Inception network contribute to the creation of a faster and more accurate architecture, particularly effective with smaller datasets (Szegedy, C et al., 2016).

DenseNet, short for Dense Convolutional Network, represents another family of CNNs employed in this study. It was conceived on the premise that convolutional networks exhibit higher accuracy and efficiency when the connections between layers closer to the input and those closer to the outputs are more direct and shorter. Within DenseNet, every layer is linked to every other layer in a feed-forward manner. The feature maps of these layers serve as inputs for subsequent layers, creating a dense connectivity pattern. These characteristics enable DenseNets to address the vanishing-gradient issue, diminish the parameter count, and promote feature reuse. Moreover, all models can operate on a 256 × 256-input size (Huang, G et al., 2017). Every DenseNet variant designed for ImageNet consists of four dense blocks incorporating 1 × 1 and 3 × 3 convolutions. The inclusion of 1 × 1 convolutions prior to each 3 × 3 convolution serves as a bottleneck mechanism, effectively diminishing the number of input feature-maps and thereby enhancing computational efficiency. Additionally, a transition layer is interposed between successive dense blocks. This transition layer comprises a sequence of operations: a 1 × 1 convolution followed by 2 × 2 average pooling, culminating in global average

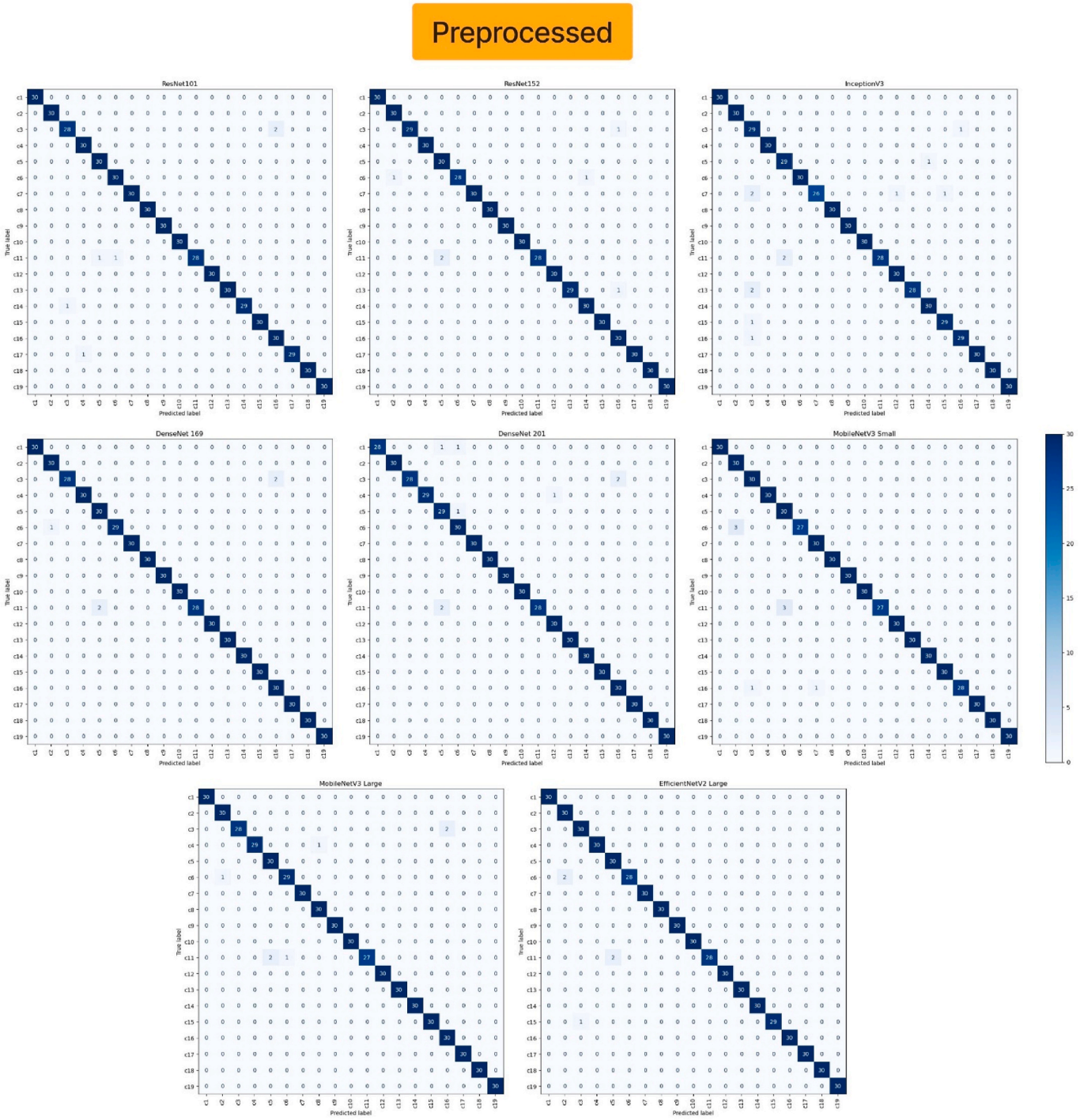


Fig. 8. Confusion matrices for all models on the preprocessed dataset.

pooling at the termination of the final dense block (Jha, N., & Pant, D. R., 2021).

MobileNet represents another suite of convolutional neural networks (CNNs) crafted by the collaborative efforts of Google AI and Brain research teams. This architecture integrates platform-aware neural architecture search (NAS) alongside NetAdapt for network exploration, thereby integrating enhancements into the network structure (Howard, A et al., 2019). In MobileNet V3, the linear bottleneck approach from MobileNet V2 was amalgamated with the inverted residual structure. This fusion is aimed at creating even more efficient layer structures, leveraging the intrinsic low-rank nature of the problem and MnasNet introduces lightweight attention modules, inspired by the squeeze-and-excitation mechanism, seamlessly integrated into the bottleneck structure, which has been built upon the MobileNetV2 structure, the layers have also been upgraded with modified swish nonlinearities (Howard, A et al., 2019). This multi-faceted strategy improves efficiency as well as contributes novel elements to the architecture, fostering a more robust and adaptive neural network design.

EfficientNet constitutes a series of models meticulously optimized for both FLOPs and parameter efficiency. These models exhibit accelerated training speeds and superior parameter efficiency compared to other models of similar complexity levels (Tan, M., & Le, Q., 2021).

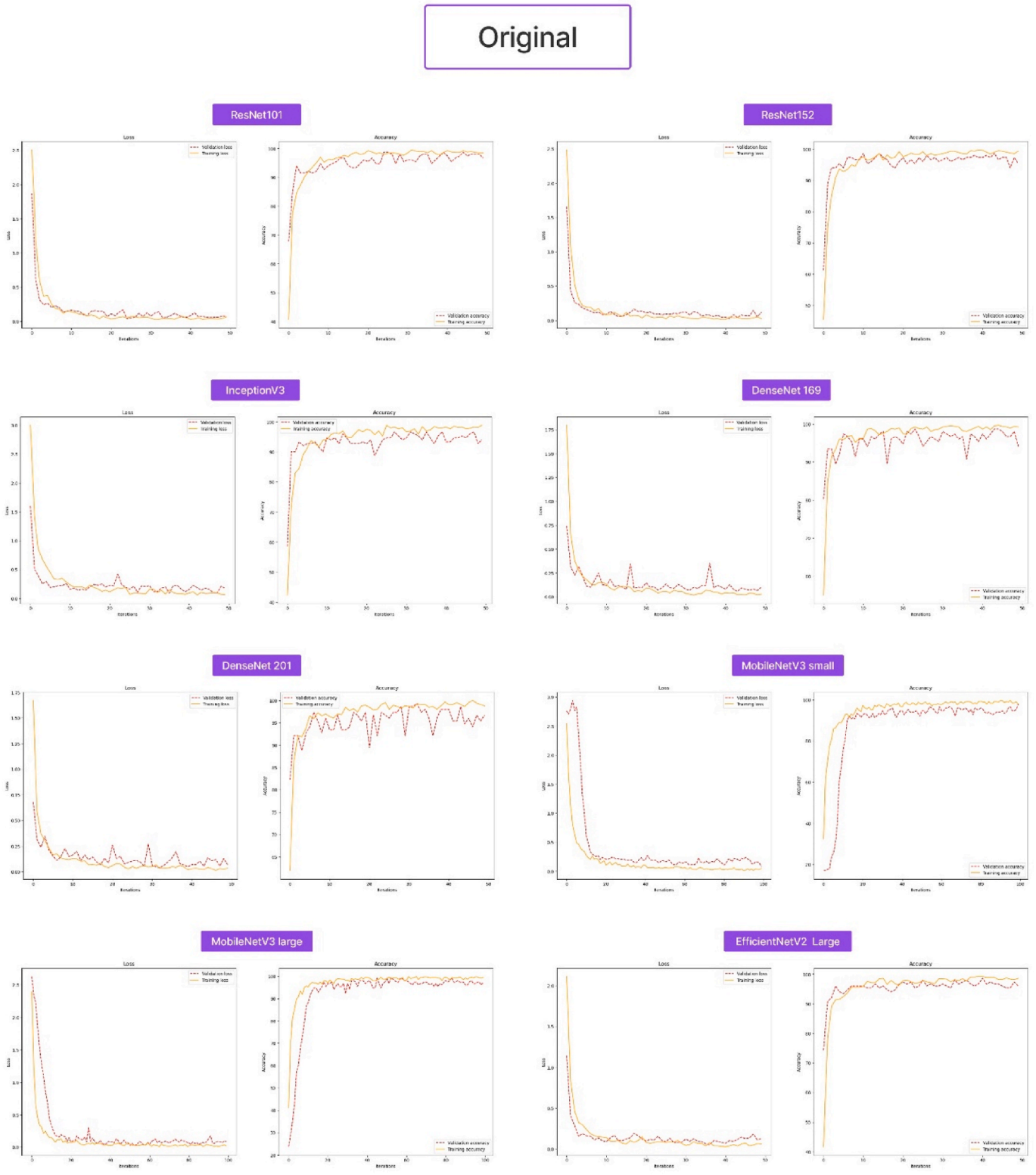


Fig. 9. Graphs illustrating the loss and accuracy of all models on original dataset.

These models are the result of a combination of training-aware neural architecture search (NAS) and scaling techniques. This joint approach aims to optimize both training speed and parameter efficiency concurrently. EfficientNetV2 heavily integrates both MBConv and fused-MBConv in its initial layers, employing smaller expansion ratios for MBConv and employing smaller 3x3 kernel sizes. To counterbalance the reduced receptive field caused by the smaller kernel size, additional layers are added, compensating for the model's large size. The input size for EfficientNetV2 is set to 224 × 224 in the study (Tan, M., & Le, Q., 2021).

The overall architecture of ResNet, Inception V3, DenseNet, MobileNet V3 small and large are shown in Fig. 4.

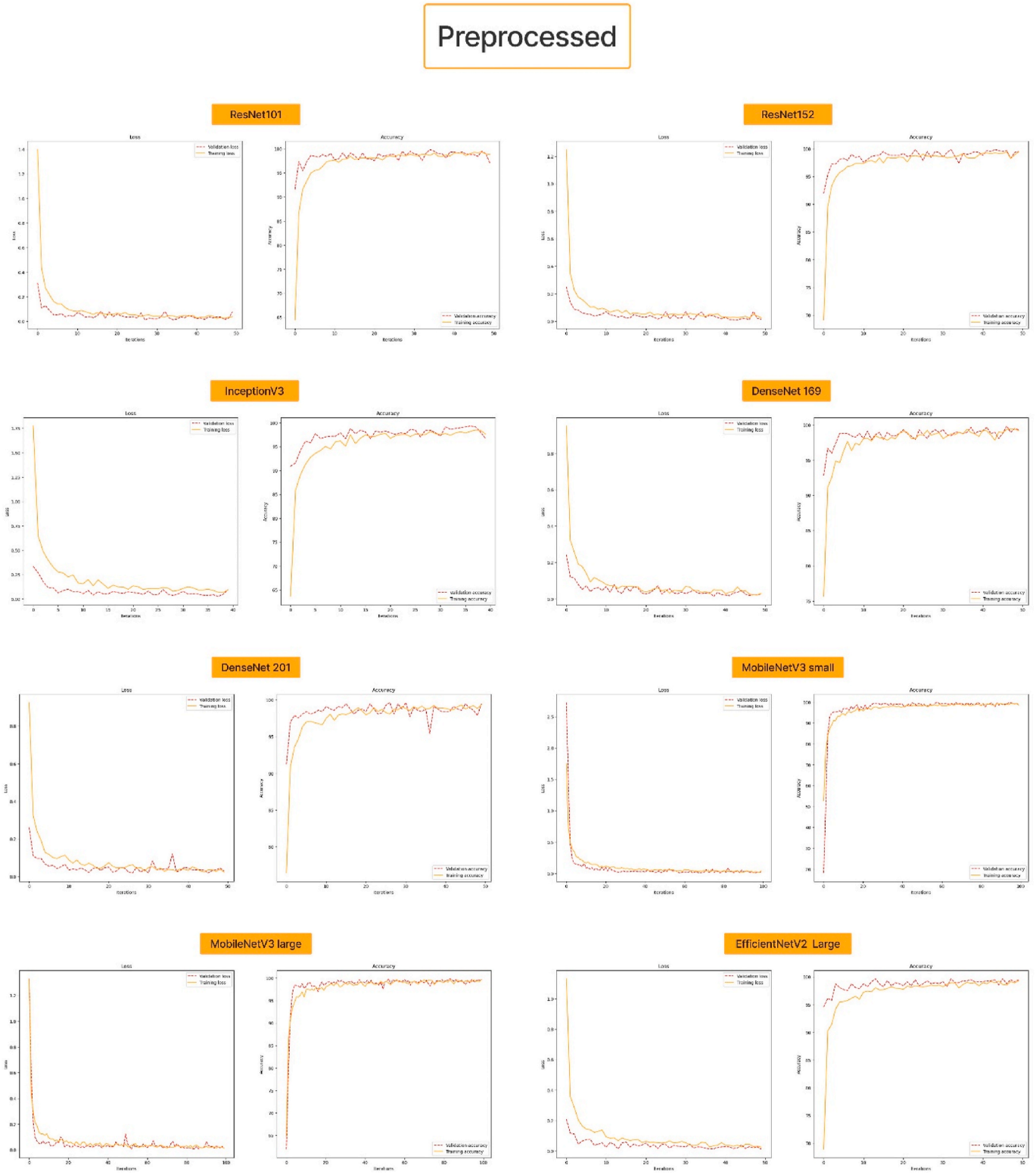


Fig. 10. Graphs illustrating the loss and accuracy for all models on the preprocessed dataset.

2.5. Fine-tuning

Due to the inherent challenges associated with a small dataset, custom deep learning model development becomes impractical due to difficulties in creating, potential overfitting risks, and time-consuming nature (Li, H et al., 2020). To tackle this challenge, the approach proposed in this study is to utilize transfer learning, a method wherein pretrained models, initially trained on extensive datasets, adapted or transferred to another related task (Tan, C et al., 2018). Transfer learning was applied to pretrained models for classification tasks, including ResNet101, ResNet152, Inception-v3, DenseNet169, DenseNet201, MobileNet-v3-small, MobileNet-v3-small, and EfficientNet-v2-large (Kornblith, S et al., 2019). These models, renowned for their exceptional performance in transfer learning tasks,

were initialized with weights obtained from the ImageNet dataset or their default weights. This avoids the utilization of random weights, which could potentially lead to undesirable local minima for the cost function. The final layer of the models was substituted with a new fully connected layer containing the same number of neurons as the number of classes in the dataset, which in this study is nineteen (19) and the complete models are retrained, this technique is known as fine-tuning, and has been widely employed for various applications (Tajbakhsh, N et al., 2016; Sharif Razavian et al., 2014). The major difference between transfer learning and fine tuning is transfer learning employs a pre-trained model as a foundation and adapts it to a new task here mainly the newly created layers are trained, while fine-tuning involves refining the parameters of a pre-trained model to suit the nuances of a new task or dataset. Fine-tuning was performed using the adaptive moment estimation (ADAM) optimizer, an algorithm for first-order gradient-based optimization known for its effectiveness in both online and non-stationary settings. ADAM combines the advantages of AdaGrad and RMSProp and works well with sparse gradients (Kingma, D. P., & Ba, J., 2014). Optimal hyperparameters for the optimizer were determined through an exhaustive search of the parameter space, recognizing the crucial role of learning parameters in fine-tuning model performance (Li, H et al., 2020). The "Grid Search" methodology was strategically employed for this purpose, systematically exploring various combinations of predefined hyperparameter values. Despite its exhaustive nature, Grid Search proves highly effective in optimizing model performance by thoroughly evaluating a spectrum of possibilities. This method is viable in terms of computational resources because of the comparably small size of the dataset and the accessibility of smaller-scale models. The parameters considered in the exploration included key aspects such as learning rate, regularization strength, and weight decay, all essential for shaping the model's behavior. By utilizing the Grid Search methodology, the study ensures a comprehensive evaluation of potential parameter configurations, enhancing the model's adaptability to the nuances of the dataset. Moreover, all models employed the ADAM optimizer along with cross-entropy loss, specifically cross-entropy loss for multi-class classification. The mathematical functions are as follows:

$$L_{cross\ entropy} = -\sum_{i}^{N} y_i \log(p_i) \tag{2}$$

$$L_{cross\ entropy\ for\ multi\ class} = -\frac{1}{N}\sum_{i}^{N}\sum_{j}^{M} y_{ij} \log\left(p_{ij}\right) \tag{3}$$

Where y_i and y_{ij} represents the true value and p_i denotes the probability value predicted by the classifier for the ith image, the variable p_{ij} represents the probability assigned by the classifier that sample i belongs to class j, N denotes the total number of rows, while M represents the total number of classes in the dataset. The optimizer works towards updating the network parameters like weights to minimize the loss function.

The deep learning models' performance was assessed using various metrics including accuracy, sensitivity, specificity, precision, and F-score. The performance indexes are as follow:

$$Accuracy = \frac{count\ of\ accurate\ predictions}{overall\ count\ of\ samples} \tag{4}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{5}$$

$$Specificity = \frac{TN}{TN + FP} \tag{6}$$

$$Precision = \frac{TP}{TN + FP} \tag{7}$$

$$F - score = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{Precision + Sensitivity} \tag{8}$$

Here, TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives. Since the dataset exhibits a balanced class distribution, accuracy serves as a straightforward and effective metric for comparing model performance. Additionally, the confusion matrix of all models was examined to identify common misclassifications among them. Moreover, the study includes accuracy and loss graphs depicting model performance during training.

3. Results and discussion

The original dataset initially contained 950 images spread across 19 classes, with each class containing 50 images. After applying preprocessing techniques, augmentation was performed, resulting in a total of 3800 images, with each of the 19 classes now comprising 200 images, up from the original 50 images. Following this, both the original and augmented datasets were divided according to the prescribed methodology. Specifically, 70% of the data was designated for training purposes, and 15% was allocated for validation testing phase, while 15% of the data was reserved for the testing phase, providing a nuanced understanding of the model's performance in real-world applications.

3.1. Dataset analysis

To visualize the high-dimensional data and assess the effectiveness of preprocessing steps, we utilized t-distributed Stochastic Neighbor Embedding (t-SNE). The t-SNE plot of the original dataset and the preprocessed dataset is presented in Fig. 5, it effectively visualizes the high-dimensional data, revealing distinct clustering patterns among the various classes labeled from C1 to C19. Each color represents a unique class, with clear separations indicating well-defined clusters. The separation will suggest whether the t-SNE algorithm has successfully captured the intrinsic structure of the data, highlighting the differences between these classes. The spread of data points within clusters varies, suggests differing levels of intra-class variability. The left plot, labeled "Original," represents the original dataset. Notably, the clusters in this plot are relatively dispersed, with several distinct groups visible, such as clusters of C2, C8, C9, and C10. Some clusters, like C1, C12, C14, C18 and C19, are not very closely packed but still maintain a degree of separation from others. However, there are notable overlaps, and some clusters are not well-defined, indicating potential noise or variability within the original data. The dispersion and overlap suggest that the raw data might contain redundant or irrelevant features, making it challenging to achieve clear separations between different classes. In contrast, the right plot, labeled "Preprocessed," showcases the dataset after applying preprocessing techniques. This visualization reveals a significant improvement in the clarity and separation of clusters. The clusters are more compact and distinctly separated from one another. For example, the C1, C2, C4, C5, C6, C8, C9, C10, C11, C12, C13, C14, C15, C17, C18 and C19 clusters are not only more cohesive but also exhibit less overlap with other clusters compared to the original data. This suggests that preprocessing has effectively reduced noise and enhanced the feature quality, thereby facilitating better discrimination between different classes. These enhancements also underscore the effectiveness of the preprocessing steps in refining the dataset, thereby facilitating better performance for downstream machine learning tasks. Overall, the t-SNE visualization highlights the critical role of data preprocessing in achieving more accurate and reliable clustering outcomes and also challenges associated with the rock classification due to intricate similarities between the classes.

3.2. Optimal hyperparameters

To optimize the training of our models, we implemented a hyperparameter tuning process using a grid search approach. Specifically, we chose the DenseNet 169 model as the foundational architecture for finding the optimal set of hyperparameters, Table 2 outlines the configuration and Grid search parameters for learning rate and weight decay. The exhaustive search involved evaluating all conceivable combinations of these crucial parameters, resulting in nine distinct sets. Each set, representing unique combinations, required approximately 2 h and 30 min for training. The accuracy achieved for each combination is succinctly summarized in Table 3. Following a comprehensive evaluation, the set displaying the highest accuracy was identified. The optimal configuration comprised of learning rate of 0.0001 and a weight decay of 1e-08. Subsequently, this hyperparameter configuration was selected for the training of all deep learning models. The final set of configurations, which are applicable to both individual model training and K-fold cross-validation, are systematically presented in Table 4. For lighter models, such as the MobileNet models, the epoch size was increased to 100 to ensure effective training.

3.3. Fine-tuning and model evaluation

Following the determination of optimal hyperparameters, Fine Tuning was applied to a diverse set of eight Deep Learning models, including ResNet101, ResNet152, Inception-v3, DenseNet169, DenseNet201, MobileNet-v3-small, MobileNet-v3-small, and EfficientNet-v2-large. The evaluation and performance comparison of these models encompassed various metrics, including accuracy, precision, sensitivity, specificity, and F-score. Given the balanced nature of the dataset, accuracy served as a robust metric for comprehensive comparison. Initially, the models were trained with the specified parameters and dataset division, and their performance on the original dataset is summarized in Table 5. DenseNet201 and InceptionV3 notably attained the highest accuracy of 98.49% on the original dataset, leading in precision, sensitivity, specificity, and F-score as well. While all models achieved accuracy surpassing 94%, it became evident that further enhancements could be achieved by incorporating preprocessing steps outlined in the Methodology. An augmented dataset was obtained, and the performance is presented in Table 6, showcasing optimal results realized on the preprocessed dataset. All models achieved accuracy exceeding 97.5%, with DenseNet169 and EfficientNetV2 large standing out with the highest accuracy of 99.12% and leading in other metrics.

Further validation and performance enhancement were conducted through K-Fold cross-validation with a fold value of five ($K = 5$) and ten ($K = 10$). During this process, the training and validation datasets were combined, whereas the test dataset remained unchanged. Each fold acted as the validation set, while the remaining folds were utilized for training. This iterative process was repeated 5 (K) times as well as 10 (K) times for 5-fold and 10-fold cross validation respectively, with each fold rotating as the validation set. The comprehensive performance of all models in 5-fold cross-validation and 10-fold cross validation is outlined in Tables 7 and 8 respectively, with all models averaging above a 98% accuracy threshold. In 5-fold cross validation MobileNet V3 large, outperformed others with the highest accuracy of 99.15%, closely followed by ResNet101, which achieved an impressive accuracy of 99.08%. DenseNet201 exhibited the least average accuracy at 98.31%, while in 10-fold cross validation too, MobileNet V3 large, outperformed others with the highest accuracy of 99.14% also leading in all the other performance metrics, it followed by MobileNet V3 small with accuracy of 98.94%. All models performed with slightly reduced accuracy by a minor margin ranging from 0.1% to 0.37% reduction on the 10-fold cross validation when compared with 5-fold cross validation, except for DenseNet201 and ResNet152 which had increase in accuracy on 10-fold cross validation, but all models still averaged above 98.24%, Inception V3 the one achieving it. The slight decrease in accuracy on the 10-fold cross validation can be associated with smaller training dataset, which increased variability in the training data across folds, as the model is trained on slightly different subsets each time. Consequently, the model might not learn as effectively from smaller training sets, resulting in a slight reduction in performance. Overall, the results showcase the robustness of the methodology with its consistent and near results on both 5-fold and 10-fold cross validation.

During training, ResNet152 and EfficientNet V2 large demanded the highest amount of training time, whereas InceptionV3 and ResNet101 exhibited the shortest training times. Comparison of various deep learning models across different variations such as original dataset, preprocessed dataset and K-Fold cross validation results for both $K = 5$ and $K = 10$ using bar plots is showcased in Fig. 6.

The confusion matrices for all models on both the original and preprocessed datasets are depicted in Figs. 7 and 8. These matrices provide insights into the classification accuracy and misclassification patterns for each model. On the original dataset the ResNet101 and ResNet152 models exhibit strong diagonal dominance, indicating high overall classification accuracy. However, some misclassifications are evident, particularly for classes C4 for the ResNet152 architecture where slight off-diagonal elements suggest confusion with C14 and C18 classes, though the ResNet101 has better accuracy and lesser misclassifications, it too shows multiple misclassifications on the original dataset. While on the preprocessed dataset the accuracy of both the models has widely increased, reaching to almost 99%, misclassification is still fairly visible, both models mainly show misclassification for the C11 rock class with both models misclassifying it as C5 rock class.

InceptionV3 also shows a strong diagonal, with minor misclassifications distributed across several classes. The model's performance is better when compared to all the architectures on the original dataset, but it has lesser accuracy on the preprocessed dataset, being the only model with reduced accuracy on the preprocessed dataset, with noticeable errors in distinguishing classes C7, C11 and C13, with mainly C7 and C13 sharing similar visual characteristics or features in the dataset visible from the t-SNE plot of the preprocessed dataset.

Both DenseNet169 and DenseNet201 models demonstrate excellent classification performance with pronounced diagonal lines in their respective confusion matrices. DenseNet201 shows slightly better performance, being the highest among all models on the original dataset, it shows fewer off-diagonal errors compared to DenseNet169, but both models misclassifying C2 class. This indicates DenseNet201's higher capacity to capture complex features due to its deeper architecture. On the preprocessed dataset, both models have an increase in accuracy, but DenseNet169 performs better, having the highest accuracy on the preprocessed dataset. Misclassification exists mainly in C11 and C3 class.

The MobileNetV3 Small model shows more pronounced misclassifications compared to the previously mentioned models on the original dataset, having the least accuracy among all the models on the original dataset. The off-diagonal elements indicate significant confusion among classes such as C2, C3, C7, C11, C17, and mainly on C16. This suggests that MobileNetV3 Small, while lightweight and efficient, might lack the necessary depth and complexity to accurately classify certain classes within this dataset but with the preprocessed dataset the accuracy increases to 98% from 94%, which shows the efficacy of the preprocessed dataset, the misclassification has been reduced to C6, C11, and C16 classes. MobileNetV3 Large shows improved performance over its smaller counterpart, with clearer diagonal and reduced off-diagonal errors. However, some classes, particularly C1, C2, C4, and C16, still exhibit notable misclassification rates, indicating areas for further refinement on the original dataset. On preprocessed dataset the required increase in accuracy is achieved, evident from its sharply defined diagonals, with main misclassification in C11 class. The confusion matrix for MobileNetV3 Large reveals fewer errors than its smaller variant on both the original and preprocessed dataset, confirming that increased model capacity positively impacts classification accuracy.

EfficientNetV2 Large demonstrate high accuracy and minimal

misclassifications though having some misclassifications, while on the preprocessed dataset it achieves the highest accuracy among all model of 99.12% just like DenseNet169 underscoring its advanced architecture and efficient scaling capabilities.

The confusion matrices highlighted the strengths and limitations of each deep learning model when applied to the original dataset. Its been predominantly observed that for the original dataset, the misclassification classes don't have a definite pattern, as misclassification changes from model to model, though classes like C2 and C16 are predominantly among the misclassified. While on the preprocessed dataset, a clear pattern of misclassification is observable, with all models having misclassified some instances of C11 as C5, also observed in the t-SNE plot of the preprocessed dataset, where some features points of C11 class lie in C5 class cluster. Also, among major misclassification common among the models trained on the preprocessed dataset is C3 class misclassified as C16, its expected from the analysis of the t-SNE plot, where its observed that C3 class feature cluster is widely scattered. Another major misclassification pattern is among C6 class misclassified as C2, though there isn't any major intersection between the two clusters. Overall, these insights are critical for guiding further model selection and improvements in preprocessing or model tuning for enhanced performance and help focus on the major misclassified classes and challenges associated with reducing it.

The training accuracy and loss graphs for both the original and preprocessed datasets are depicted in Figs. 9 and 10, respectively. These graphs highlight that, compared to the original dataset with multiple dips, all models on the preprocessed dataset converged faster without overfitting, albeit requiring a longer training duration. In summary, the study achieved remarkable results, with multiple models attaining accuracy levels exceeding 99% on the preprocessed dataset and in K-Fold cross-validation. This underscores the robustness and effectiveness of the proposed methodologies, validating the success of the study in advancing the understanding of rock types.

4. Conclusion

In conclusion, this study presents a comprehensive approach to tackling the complex challenges associated with geological image segmentation using advanced deep learning methodologies. Leveraging recent advancements in image segmentation, particularly in medical imaging and object recognition, our tailored methodology proved effective in capturing a diverse range of geological formations. The establishment of a dataset containing 950 images of 19 distinct rock types, each documented with 50 samples, laid the foundation for the subsequent success of our approach. The two crucial phases of dataset preprocessing and deep learning model application, employing ResNets, Inception V3, DenseNets, MobileNets V3, and EfficientNet V2 large, demonstrated exceptional performance metrics. Optimal hyperparameter tuning and fine-tuning of eight deep learning models, particularly DenseNet201 and InceptionV3, showcased remarkable accuracy levels, exceeding 99% on the preprocessed dataset and in K-Fold cross-validation. Notably, MobileNet V3 large exhibited the highest accuracy of 99.15% in the cross-validation process. The incorporation of preprocessing steps proved pivotal in enhancing the models' performance, promoting faster convergence without overfitting. Overall, the methodologies used in this research yielded outstanding results, confirming the efficacy of our approach in advancing the understanding of rock types through precise image segmentation. This research makes a significant contribution to the broader field of geological analysis by offering a robust framework for accurate and efficient classification across diverse geological formations.

Code availability statement

The code developed for this research was created using Python and open-source libraries including PyTorch, Matplotlib, and others. The complete mention of libraries and their version is available on the GitHub repository. The code is fully compliant with open-source principles. The following details are provided regarding the availability of the code:

Name and Version of the Code: version 1.

License Type: MIT License.

Link to Public Repository: https://github.com/Phantom-fs/Rock-Type-Classification/

CRediT authorship contribution statement

Amit Kumar Gupta: Writing – original draft, Methodology, Investigation, Conceptualization. **Priya Mathur:** Writing – review & editing, Visualization, Conceptualization. **Farhan Sheth:** Writing – review & editing, Methodology, Investigation. **Carlos M. Travieso-Gonzalez:** Writing – original draft, Supervision, Data curation. **Sandeep Chaurasia:** Visualization, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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